Properties of equality and congruence

1 Complete the following statements.

a The addition property of equality states that if $a = b$, then $a + c =$ ▯.

b The multiplication property of equality states that if $a = b$ and $c \neq 0$, then $ac =$ ▯.

c The symmetric property of equality states that if $AB = CD$, then $CD =$ ▯.

2 By which property of equality are $9x = -27$ and $\frac{9x}{9} = \frac{-27}{9}$ equivalent equations?

3 For each of the following true expressions, determine which property of equality or congruence is illustrated.

a If $AB = CD$, then $CD = AB$

b If $AB = CD$, then $AB + 3 = CD + 3$

c If $\overline{AB} \cong \overline{CD}$ and $\overline{CD} \cong \overline{EF}$, then $\overline{AB} \cong \overline{EF}$

4 If $\overline{PR}$ is a line segment, determine whether the following statements illustrate the reflexive property of congruence.

a $PR \cong RP$

b $PR = PR$

c $\overline{PR} \cong \overline{PR}$

d $\overline{PR} = \overline{RP}$

5 Consider the following true statement:

$$AB = 12$$

Determine whether the following are valid deductions from this statement.

a $\frac{AB}{2} = 6$

b $AB + 5 = 17$

c $2AB = 6$

d $12 = AB$

6 Consider the following true statement:

$$AB + CD = 32$$

Determine whether the following is a valid deduction from this statement using only a single property of equality.

a $BA + DC = 16 + 16$

b $AB + AB = 32 + CD$

c $2AB = 64 - CD$

d $32 = AB + CD$

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7 For each of the following arguments, determine which property of equality is illustrated.

a Given: $m\angle ABC = m\angle DEF$ and $m\angle DEF = 75^\circ$
Conclusion: $m\angle ABC = 75^\circ$

b Given: $PR = QD$
Conclusion: $PR \times PR = PR \times QD$

c Given: $AB = 16$
Conclusion: $AB - 5 = 11$

8 Consider the following argument.
Given: $\angle TSR \cong \angle QSP$ and $\angle TSR \cong \angle APC$
Conclusion: $\angle QSP \cong \angle APC$
Determine whether the argument relies upon each of the following properties.

a The reflexive property of equality.

b The symmetric property of congruence.

c The symmetric property of equality.

d The reflexive property of congruence.

e The transitive property of congruence.

f The transitive property of equality.

Additional questions

9 Copy and complete the table for the properties of equality:

Property of Equality	Example
Reflexive property of equality	$a = a$
Symmetric property of equality	▯
Transitive property of equality	▯
Addition property of equality	▯
▯	If $a = b$, then $a - c = b - c$.
▯	If $a = b$, then $a \times c = b \times c$.
Division property of equality	▯
Substitution property of equality	▯

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10 For each of the following pairs of steps, determine which property or converse property of equality is used to get from Step 1 to Step 2.

a

1. $\angle A = \angle B$
2. $\angle A + \angle C = \angle B + \angle C$

b

1. $3x = 24$
2. $x = 8$

c

1. $p + r = q + r$
2. $p = q$

d

1. $\angle X - \angle Z = \angle Y - \angle Z$
2. $\angle X = \angle Y$

11 For each of the following sets of steps, determine which property or converse property of equality is used to get to Step 3.

a

1. $x = 23$
2. $y = 23$
3. $x = y$

b

1. $x = y + 2$
2. $2x - 3y = -4$
3. $2(y + 2) - 3y = -4$

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